

```
In [33]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib # Add this line

print("✅ Environment setup successful!")
print(f"pandas: {pd.__version__}")
print(f"numpy: {np.__version__}")
print(f"matplotlib: {matplotlib.__version__}")
print(f"seaborn: {sns.__version__}")
```

✅ Environment setup successful!
pandas: 3.0.0
numpy: 2.4.2
matplotlib: 3.10.8
seaborn: 0.13.2

```
In [34]: # Load the Titanic dataset
df = pd.read_csv('../datasets/titanic.csv')

# Display first few rows
print(f"Dataset shape: {df.shape[0]} rows, {df.shape[1]} columns")
print("\nFirst x rows:")
df.head(25)
```

Dataset shape: 891 rows, 12 columns

First x rows:

```
Out[34]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0
5	6	0	3	Moran, Mr. James	male	NaN	0	0

6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0
10	11	1	3	Sandstrom, Miss. Marguerite Rut	female	4.0	1	1
11	12	1	1	Bonnell, Miss. Elizabeth	female	58.0	0	0
12	13	0	3	Saundercock, Mr. William Henry	male	20.0	0	0
13	14	0	3	Andersson, Mr. Anders Johan	male	39.0	1	5
14	15	0	3	Vestrom, Miss. Hulda Amanda Adolfina	female	14.0	0	0
15	16	1	2	Hewlett, Mrs. (Mary D Kingcome)	female	55.0	0	0
16	17	0	3	Rice, Master. Eugene	male	2.0	4	1
17	18	1	2	Williams, Mr. Charles Eugene	male	NaN	0	0
18	19	0	3	Vander Planke, Mrs. Julius (Emelia Maria Vande...	female	31.0	1	0

19	20	1	3	Masselmani, Mrs. Fatima	female	NaN	0	0
20	21	0	2	Fynney, Mr. Joseph J	male	35.0	0	0
21	22	1	2	Beesley, Mr. Lawrence	male	34.0	0	0
22	23	1	3	McGowan, Miss. Anna "Annie"	female	15.0	0	0
23	24	1	1	Sloper, Mr. William Thompson	male	28.0	0	0
24	25	0	3	Palsson, Miss. Torborg Danira	female	8.0	3	1

In [35]: `print(df.info())` # Data types and missing values

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null    int64
1   Survived        891 non-null    int64
2   Pclass          891 non-null    int64
3   Name            891 non-null    str
4   Sex             891 non-null    str
5   Age             714 non-null    float64
6   SibSp           891 non-null    int64
7   Parch           891 non-null    int64
8   Ticket          891 non-null    str
9   Fare            891 non-null    float64
10  Cabin           204 non-null    str
11  Embarked        889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
None
```

In [36]: `print(df.describe().round(2))`

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.00	891.00	891.00	714.00	891.00	891.00	891.00
mean	446.00	0.38	2.31	29.70	0.52	0.38	32.20
std	257.35	0.49	0.84	14.53	1.10	0.81	49.69
min	1.00	0.00	1.00	0.42	0.00	0.00	0.00
25%	223.50	0.00	2.00	20.12	0.00	0.00	7.91
50%	446.00	0.00	3.00	28.00	0.00	0.00	14.45
75%	668.50	1.00	3.00	38.00	1.00	0.00	31.00
max	891.00	1.00	3.00	80.00	8.00	6.00	512.33

In [37]: *# Calculate then print specific statistics*

```
mean_age = df['Age'].mean()
print(f"Mean age: {mean_age:.2f}")

median_age = df['Age'].median()
print(f"Median age: {median_age:.2f}")

std_age = df['Age'].std()
print(f"Std Age: {std_age:.2f}")
```

Mean age: 29.70

Median age: 28.00

Std Age: 14.53

In [38]: *# Deciles from 0 to 100% (including min and max)*

```
print(df['Age'].quantile([i/10 for i in range(11)]))
# This gives: 0.0, 0.1, 0.2, 0.3, ... 0.9, 1.0
```

```
0.0    0.42
0.1    14.00
0.2    19.00
0.3    22.00
0.4    25.00
0.5    28.00
0.6    31.80
0.7    36.00
0.8    41.00
0.9    50.00
1.0    80.00
```

Name: Age, dtype: float64

In [39]: *# Percentiles*

```
print(df['Age'].quantile([0.25, 0.5, 0.75, 0.95, 0.99]))
```

```
0.25    20.125
0.50    28.000
0.75    38.000
0.95    56.000
0.99    65.870
```

Name: Age, dtype: float64

In [40]: *# Survival rate*

```
survival_rate = df['Survived'].mean()
print(f"Overall survival rate: {survival_rate:.2%}")
```

Overall survival rate: 38.38%

```
In [42]: # Group statistics
print(df.groupby('Sex')['Survived'].mean())
print(df.groupby('Pclass')['Survived'].mean())
```

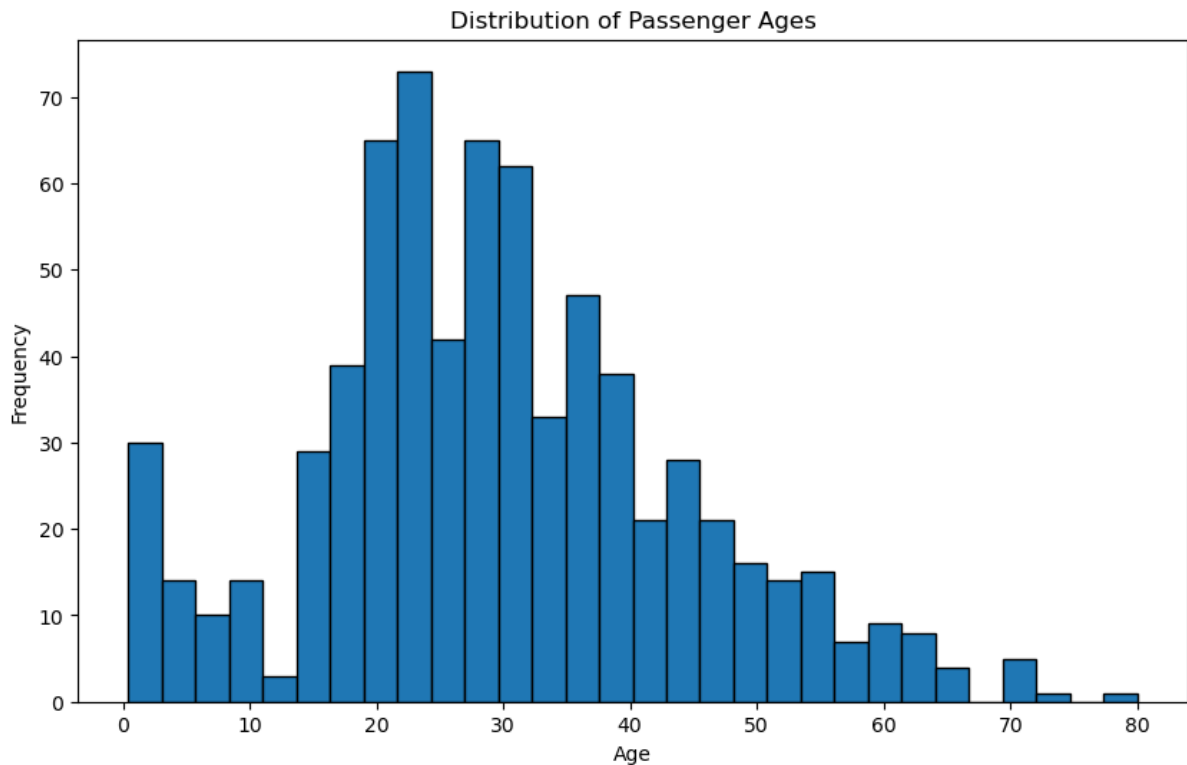
```
Sex
female    0.742038
male      0.188908
Name: Survived, dtype: float64
Pclass
1         0.629630
2         0.472826
3         0.242363
Name: Survived, dtype: float64
```

```
In [43]: std_age = df['Age'].std()
var_age = df['Age'].var()

print (f"stg_age: {std_age:.2f}") # standard deviation
print(f"variance_age: {var_age:.2f}") # variance (std squared)
```

```
stg_age: 14.53
variance_age: 211.02
```

```
In [44]: # Visualizations
plt.figure(figsize=(10, 6))
plt.hist(df['Age'].dropna(), bins=30, edgecolor='black')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('Distribution of Passenger Ages')
plt.show()
```



```
In [45]: age_skew = df['Age'].skew()
print (f"age_skew = {age_skew:.2f}")
```

age_skew = 0.39

```
In [46]: print(f"Mean:    {df['Age'].mean():.2f}")
print(f"Median:  {df['Age'].median():.2f}")
print(f"Mode:    {df['Age'].mode()[0]:.2f}")
print(f"Std:      {df['Age'].std():.2f}")
```

Mean: 29.70
Median: 28.00
Mode: 24.00
Std: 14.53